

Assignment 4

Start Assignment

- Due Monday by 11:59pm
- Points 100
- Submitting a file upload

No additional details were added for this assignment.

HW4 Rubric - Object Detection (SP26) FINAL

Criteria	Ratings				Pts
Report/YOLO: mAP, Kaggle mAP, test loss Report all three: mAP value, Kaggle mAP, and test loss for YOLO	2 to >1.0 pts Full Marks All three reported	1 to >0.0 pts Partial (1 pts) One missing	0 pts No Credit Two or three missing		2 pts
Report/YOLO: Hyperparameter search description Describe which hyperparameter sets were tried (including default) and why.	3 to >2.0 pts Full Marks At least 2 sets + reasoning	2 to >1.0 pts Partial (2 pts) At least 2 sets, no reasoning	1 to >0.0 pts Partial (1 pts) Only final hyperparameter reported	0 pts No Credit Missing	3 pts
Report/YOLO: Hyperparameter search observations Observations about model behavior per hyperparameter set tried.	3 to >1.0 pts Full Marks Reasonable observations per set	1 to >0.0 pts Partial (1 pts) Insufficient observations	0 pts No Credit Missing		3 pts
Report/YOLO: What didn't work + fixes What didn't work and how it was fixed. OK if 'everything worked fine'.	1 to >0.0 pts Full Marks Present (or 'everything worked')	0 pts No Credit Missing			1 pts
Report/YOLO: Sample images At least one sample image with detections. OK if only one image.	1 to >0.0 pts Full Marks At least one image	0 pts No Credit Missing			1 pts

Criteria	Ratings				Pts
Report/DETR: mAP, Kaggle mAP, test loss Report all three: mAP value, Kaggle mAP, and test loss for DETR.	2 to >1.0 pts Full Marks All three reported	1 to >0.0 pts Partial (1 pts) One missing	0 pts No Credit Two or three missing		2 pts
Report/DETR: Hyperparameter search description Describe which hyperparameter sets were tried (including default) and why.	3 to >2.0 pts Full Marks At least 2 sets + reasoning	2 to >1.0 pts Partial (2 pts) At least 2 sets, no reasoning	1 to >0.0 pts Partial (1 pts) Only final hyperparameter reported	0 pts No Credit Missing	3 pts
Report/DETR: Hyperparameter search observations Observations about model behavior per hyperparameter set tried.	3 to >1.0 pts Full Marks Reasonable observations per set	1 to >0.0 pts Partial (1 pts) Insufficient observations	0 pts No Credit Missing		3 pts
Report/DETR: What didn't work + fixes What didn't work and how it was fixed. OK if 'everything worked fine'.	1 to >0.0 pts Full Marks Present (or 'everything worked')	0 pts No Credit Missing			1 pts
Report/DETR: Sample images At least one sample image with detections. OK if only one image.	1 to >0.0 pts Full Marks At least one image	0 pts No Credit Missing			1 pts

Criteria	Ratings						Pts
Performance/YOLO: mAP (baseline 53%) Exact score = average of public and private Kaggle leaderboard scores. Baseline: 53% mAP. $\geq 52.5\%$ -> 15 pts; $\geq 51.5\%$ -> 14 pts; $\geq 50.5\%$ -> 13 pts; -1 pt per 1% absolute decrease in mAP. -5 pts for missing Kaggle submission. -5 pts if num_epochs != 15.	15 to >14.0 pts Full Marks mAP $\geq 52.5\%$	14 to >13.0 pts mAP $\geq 51.5\%$ mAP $\geq 51.5\%$	13 to >10.0 pts mAP $\geq 50.5\%$ mAP $\geq 50.5\%$	10 to >5.0 pts mAP $\sim 47.5\%$ (-5 below full) mAP $\sim 47.5\%$ (-5 below full)	5 to >0.0 pts mAP $\sim 42.5\%$ (-10 below full) mAP $\sim 42.5\%$ (-10 below full)	0 pts No Credit mAP $\leq 37.5\%$ or missing	15 pts
Performance/DETR: mAP (baseline 58%) Exact score = average of public and private Kaggle leaderboard scores. Baseline: 58% mAP. $\geq 57.5\%$ -> 15 pts; $\geq 56.5\%$ -> 14 pts; $\geq 55.5\%$ -> 13 pts; -1 pt per 1% absolute decrease in mAP. -5 pts for missing Kaggle submission. -5 pts if num_epochs != 15.	15 to >14.0 pts Full Marks mAP $\geq 57.5\%$	14 to >13.0 pts mAP $\geq 56.5\%$ mAP $\geq 56.5\%$	13 to >10.0 pts mAP $\geq 55.5\%$ mAP $\geq 55.5\%$	10 to >5.0 pts mAP $\sim 52.5\%$ (-5 below full) mAP $\sim 52.5\%$ (-5 below full)	5 to >0.0 pts mAP $\sim 47.5\%$ (-10 below full) mAP $\sim 47.5\%$ (-10 below full)	0 pts No Credit mAP $\leq 42.5\%$ or missing	15 pts
Code/YOLO: split_prediction_tensor Sub: (a) slice first B*5 channels as boxes, remainder as classes [1 pt]; (b) reshape to boxes [N,S,S,B,5] and classes [N,S,S,C] [1 pt].	2 to >1.0 pts Full Marks Full Marks (both sub-items)		1 to >0.0 pts Partial (1 pts) Partial (1 sub-item)		0 pts No Credit No code / neither		2 pts

Criteria	Ratings					Pts
Code/YOLO: xywh_to_xyxy Sub: (a) center = (xy + col_row)/grid_size [1]; (b) half wh [1]; (c) corners top-left/bottom-right [1]; (d) [x1,y1,x2,y2] order [1].	4 to >3.0 pts Full Marks Full Marks (all 4 sub-items)	3 to >2.0 pts Partial (3 pts) Partial (3 of 4)	2 to >1.0 pts Partial (2 pts) Partial (2 of 4)	1 to >0.0 pts Partial (1 pts) Partial (1 of 4)	0 pts No Credit No code	4 pts
Code/YOLO: choose_responsible_box Sub: (a) has_object_mask filtering [2]; (b) empty-object edge case [1]; (c) xywh_to_xyxy for IoU input [2]; (d) IoU matrix diagonal selection [2]; (e) correct return shapes [num_cells,5],[num_cells,1], [num_cells] [2].	9 to >7.0 pts Full Marks Full Marks (all sub-items)	7 to >5.0 pts Partial (7 pts) Mostly correct (7 of 9)	5 to >3.0 pts Partial (5 pts) Partial (5 of 9)	3 to >0.0 pts Partial (3 pts) Partial (3 of 9)	0 pts No Credit No code	9 pts
Code/YOLO: build_responsible_mask Sub: (a) no-object-cells edge case [1]; (b) correct mask shape [N,S,S,B] [2].	3 to >2.0 pts Full Marks Full Marks (both sub-items)	2 to >1.0 pts Partial (2 pts) Partial (shape only, missing edge case)	1 to >0.0 pts Partial (1 pts) Partial (edge case only)	0 pts No Credit No code		3 pts
Code/YOLO: regression_xy_loss Sub: (a) extract xy values [1]; (b) SSE on (x,y) offsets [1].	2 to >1.0 pts Full Marks Full Marks (both sub-items)	1 to >0.0 pts Partial (1 pts) Partial (1 of 2)		0 pts No Credit No code		2 pts

Criteria	Ratings						Pts
Code/YOLO: regression_wh_loss Sub: (a) extract wh [1]; (b) sqrt instead of raw sizes [1]; (c) clamp before sqrt [1]; (d) SSE [1]. CRITICAL: missing sqrt is the 'major deduction' Kulbir flagged with Kiet.	4 to >3.0 pts Full Marks Full Marks (all 4 sub-items)	3 to >2.0 pts Partial (3 pts) Partial (3 of 4)	2 to >1.0 pts Partial (2 pts) Partial (2 of 4)	1 to >0.0 pts Partial (1 pts) Partial (1 of 4)	0 pts No Credit No code		4 pts
Code/YOLO: object_confidence_loss Sub: (a) extract conf values [1]; (b) IoU as confidence target (NOT 1.0) [1]; (c) detach IoU [1]; (d) SSE [1].	4 to >3.0 pts Full Marks Full Marks (all 4 sub-items)	3 to >2.0 pts Partial (3 pts) Partial (3 of 4)	2 to >1.0 pts Partial (2 pts) Partial (2 of 4)	1 to >0.0 pts Partial (1 pts) Partial (1 of 4)	0 pts No Credit No code		4 pts
Code/YOLO: no_object_confidence_loss Sub: (a) extract conf [1]; (b) use ~responsible_mask for no-object predictors [1]; (c) target = 0 [1]; (d) SSE [1].	4 to >3.0 pts Full Marks Full Marks (all 4 sub-items)	3 to >2.0 pts Partial (3 pts) Partial (3 of 4)	2 to >1.0 pts Partial (2 pts) Partial (2 of 4)	1 to >0.0 pts Partial (1 pts) Partial (1 of 4)	0 pts No Credit No code		4 pts
Code/YOLO: class_probability_loss Sub: (a) use has_object_mask for object cells only [2]; (b) SSE on class probs [1].	3 to >2.0 pts Full Marks Full Marks (both sub-items)	2 to >1.0 pts Partial (2 pts) Partial (mask only)	1 to >0.0 pts Partial (1 pts) Partial (SSE only)	0 pts No Credit No code			3 pts
Code/YOLO: forward Sub: (a) correct order and use of helpers [1]; (b) lambda_coord	5 to >4.0 pts Full Marks	4 to >3.0 pts Partial (4 pts)	3 to >2.0 pts Partial (3 pts)	2 to >1.0 pts Partial (2 pts)	1 to >0.0 pts Partial (1 pts)	0 pts No Credit	5 pts

Criteria	Ratings						Pts
weighting [1]; (c) lambda_noobj weighting [1]; (d) normalize by batch_size [1]; (e) correct dict return [1].	Full Marks (all 5 sub-items)	Partial (4 of 5)	Partial (3 of 5)	Partial (2 of 5)	Partial (1 of 5)	No code	
Code/DETR: SimpleDETR __init__ 5 modules: input_proj 1x1 Conv2d (no hardcode [1] + kernel [1]) [2]; transformer (-0.5 per missing arg) [3]; query_embed [1]; class_embed [1]; bbox_embed MLP [1].	8 to >6.0 pts Full Marks Full Marks (all 5 modules correct)	6 to >4.0 pts Partial (6 pts) Partial (4 of 5)	4 to >2.0 pts Partial (4 pts) Partial (3 of 5)	2 to >0.0 pts Partial (2 pts) Partial (2 of 5)	0 pts No Credit No code		8 pts
Code/DETR: SimpleDETR forward Sub: backbone features [1]; input_proj [1]; positional encoding [1]; flatten src+pos+mask [2]; expand query [1]; feed src+tgt w/ src_key_padding_mask + memory_key_padding_mask [3]; class logits [1]; bbox MLP + sigmoid [2].	12 to >9.0 pts Full Marks Full Marks	9 to >6.0 pts Partial (9 pts) Partial (minor issue, ~9 of 12)	6 to >3.0 pts Partial (6 pts) Partial (moderate, ~6 of 12)	3 to >0.0 pts Partial (3 pts) Partial (major, ~3 of 12)	0 pts No Credit No code		12 pts
Code/DETR: loss_labels Sub: matched indices via __get_src_permutation_idx [2]; GT label tensor via zip [2] + indices[idx] [1] [2]; init all queries to no-object [2]; overwrite matched w/ GT [2]; CE loss	14 to >10.0 pts Full Marks Full Marks	10 to >6.0 pts Partial (10 pts) Partial (~10 of 14)	6 to >3.0 pts Partial (6 pts) Partial (~6 of 14)	3 to >0.0 pts Partial (3 pts) Partial (~3 of 14)	0 pts No Credit No code		14 pts

Criteria	Ratings						Pts
(1) + transpose (1) + empty_weight (2) = [4].							
Code/DETR: loss_boxes Sub: matched indices [2]; GT box tensor via zip+indices [4]; L1 sum (2) + normalize (2) = [4]; cxcywh->xyxy [2]; pairwise GloU [2]; complementary GloU diagonal [2]; normalize [2].	18 to >14.0 pts Full Marks Full Marks	14 to >10.0 pts Partial (14 pts) Partial (~14 of 18)	10 to >6.0 pts Partial (10 pts) Partial (~10 of 18)	6 to >3.0 pts Partial (6 pts) Partial (~6 of 18)	3 to >0.0 pts Partial (3 pts) Partial (~3 of 18)	0 pts No Credit No code	18 pts
Code/DETR: DETRSetCriterion forward Sub: Hungarian matcher [2]; num_boxes = TARGET boxes [2]; invoke loss_labels [1]; invoke loss_boxes [1]; combine + return dict [2].	8 to >6.0 pts Full Marks Full Marks	6 to >4.0 pts Partial (6 pts) Partial (6 of 8)	4 to >2.0 pts Partial (4 pts) Partial (4 of 8)	2 to >0.0 pts Partial (2 pts) Partial (2 of 8)	0 pts No Credit No code		8 pts
EC: Alternative backbone (capped 5) Must differ from all assignment or other-EC backbones. Each backbone +2; each discussion +1. Cap 5.	5 to >3.0 pts Full Marks Full cap (2+ backbones with discussion)	3 to >2.0 pts Partial (3 pts) Partial (one backbone + discussion)	2 to >0.0 pts Partial (2 pts) Minimal (one backbone, no discussion)	0 pts No Credit None			5 pts
EC: Alternative architecture (capped 8)	8 to >4.0 pts Full Marks	4 to >3.0 pts Partial (4 pts)	3 to >0.0 pts Partial (3 pts)	0 pts No Credit			8 pts

Criteria	Ratings					Pts
Must differ from all assignment or other-EC architectures. Each architecture +3; each discussion +1. Cap 8.	Full cap (2+ architectures with discussion)	Partial (one architecture + discussion)	Minimal (one architecture, no discussion)	None		
EC: Alternative loss (capped 5) Must differ from all assignment or other-EC losses. Each loss +2; each discussion +1. Cap 5.	5 to >3.0 pts Full Marks Full cap (2+ losses with discussion)	3 to >2.0 pts Partial (3 pts) Partial (one loss + discussion)	2 to >0.0 pts Partial (2 pts) Minimal (one loss, no discussion)	0 pts No Credit None		5 pts
EC: Incorporating anchors in YOLO Implementation [3] + Discussion [1].	4 to >3.0 pts Full Marks Full Marks (impl + disc)	3 to >1.0 pts Partial (3 pts) Impl only	1 to >0.0 pts Partial (1 pts) Disc only	0 pts No Credit None		4 pts
EC: Visualization YOLO: viz [1] + disc [1]. DETR: viz [1] + disc [1].	4 to >3.0 pts Full Marks Full Marks (YOLO + DETR viz + disc)	3 to >2.0 pts Partial (3 pts) Three of four sub-items	2 to >1.0 pts Partial (2 pts) Two of four sub-items	1 to >0.0 pts Partial (1 pts) One of four sub-items	0 pts No Credit None	4 pts
EC: Direct comparisons between similarly-sized backbones/models YOLO+ResNet34 [1] + DETR+ResNet34 [1] + Comprehensive comparison [2].	4 to >3.0 pts Full Marks Full Marks (both + comprehensive)	3 to >2.0 pts Partial (3 pts) Both backbones, partial comparison	2 to >1.0 pts Partial (2 pts) One backbone + comparison	1 to >0.0 pts Partial (1 pts) One backbone only	0 pts No Credit None	4 pts
Total Points: 180						